Optimizing Machine Learning Models for Smoking Status Prediction

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1. Objectives

- To predict the smoking status of individuals based on biological health indicators.
- To apply metaheuristic optimization techniques to enhance classification models.
- To compare performance before and after hyperparameter tuning.
- To utilize GWO and PSOGSA algorithms for optimal accuracy.

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2. Problem Description

Predicting smoking status using biological markers such as hemoglobin, cholesterol, and liver enzymes is challenging due to high correlation among features. Traditional machine learning models rely on default parameters, which may not yield optimal results.

This project replaces manual tuning with intelligent optimization techniques to find the best hyperparameters, improving model accuracy and reliability.

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3. Mathematical Formulation

Decision Variables

- SVM: X = [C, gamma]
- XGBoost: X = [n_estimators, max_depth, eta]

Objective Function

Maximize:

$f(X) = \text{Accuracy}(X)$

Subject to:

$X \in [LB, UB]$

Constraints

- $0.1 \le C \le 100$
- $0.0001 \le \text{gamma} \le 1$
- $50 \le \text{n\_estimators} \le 300$
- $2 \le \text{max\_depth} \le 10$
- $0.01 \le \text{eta} \le 0.3$

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4. Optimization Algorithms Formulation

This study focuses on two selected optimization algorithms: Grey Wolf Optimizer (GWO) and Hybrid PSOGSA. These algorithms were chosen due to their strong performance in handling both low-dimensional (SVM) and high-dimensional (XGBoost) optimization problems. While other algorithms such as PSO and GSA exist, they were not included in the final implementation to maintain computational efficiency and focus on the most effective approaches for this problem.

4.1 Grey Wolf Optimizer (GWO)

D = |C * X_p - X|
X(t+1) = X_p - A * D
Objective:
Minimize: f(X) = 1 - Accuracy
Constraints:
X ∈ [LB, UB]

4.2 Hybrid PSOGSA

Velocity Update:
V_i(t+1) = wV_i(t) + c1*r1*(Pbest_i - X_i(t)) + a_i(t)
Position Update:
X_i(t+1) = X_i(t) + V_i(t+1)
Objective:
Minimize: f(X) = 1 - Accuracy
Constraints:
X ∈ [LB, UB]

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4. Methodology

A. Grey Wolf Optimizer (GWO)
GWO is used to optimize SVM parameters by balancing exploration and exploitation.
Steps:

1. Initialize wolves (solutions)

2. Identify Alpha, Beta, Delta
3. Update positions based on leaders
4. Repeat until convergence

B. Hybrid PSOGSA

PSOGSA combines PSO and GSA for optimizing XGBoost.

Steps:

1. Initialize particles
2. Compute gravitational forces
3. Update velocity using PSO rules
4. Update positions

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5. Results

Model	Algorithm	Accuracy Before	Accuracy After
SVM	GWO	76%	78%
XGBoost	PSOGSA	78%	81%

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6. Discussion

The results show that optimization significantly improves model performance. GWO improved SVM decision boundaries, while PSOGSA effectively handled the complex search space of XGBoost.

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7. Conclusion

Optimization techniques enhance accuracy and reliability of machine learning models. Hybrid algorithms like PSOGSA are especially effective for complex models.

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8. References

1. Grey Wolf Optimizer (2014)
2. PSOGSA Algorithm (2010)
3. XGBoost Paper (2016)